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Derivative Gaussian Process for scRNAseq Data using Combination Covariance Functions

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Abstract

The goal of this thesis is to investigate the use of combination covariance functions in multi-output derivative Gaussian processes (GPs) for modelling single-cell RNA sequencing (scRNAseq) data. Recent work introduced derivative GPs for simultaneously modelling expression values and their RNA velocities over a pseudotime. However, biological processes can be highly diverse and exhibit too complex dynamics for standard covariance functions. We aim to investigate whether combination covariance functions are capable of addressing this issue. Hence, investigating these combination covariance functions is crucial for the development of robust models for scRNAseq data. Here, we explore the additive combination of the commonly used squared exponential (SQE) and Matérn 3/2 covariance functions and propose a new variant of it. To quantify uncertainty in estimated model hyperparameters, we use Bayesian inference for model fitting. To study the effects of the proposed covariance functions, we performed an extensive simulation study to evaluate the ground truth hyperparameter recovery and predictive performance. Furthermore, two case studies were performed to evaluate the covariance functions on real scRNAseq datasets. Through our analysis, we confirm that the commonly used SQE covariance function is unable to capture complex data. Based on the case study we cannot confirm that the additive combination of covariance functions produces better-suited models for scRNAseq data. Given the similar performance of standard covariance functions, it raises the question of whether developing combination covariance functions is indeed necessary, increasing model complexity while providing only a limited understanding of the underlying biological process.

Zusammenfassung

Das Ziel dieser Arbeit ist es, den Einsatz von kombinierten Kovarianzfunktionen in mehrdimensionalen Gauß-Prozessen (GPs) mit integrierter Ableitung zur Modellierung von single-cell RNA Sequenzier-Daten (scRNAseq Daten) zu untersuchen. Aktuelle Publikationen haben GPs mit integrierter Ableitung zur gleichzeitigen Modellierung von Expressionsdaten und dazugehörigen RNA-velocities und Pseudotimes eingeführt. Biologische Prozesse können jedoch äußerst vielfältig sein und zu komplexe Dynamiken hervorbringen, sodass diese nicht mit Standard-Kovarianzfunktionen modelliert werden können. Wir möchten dementsprechend untersuchen, ob eine Kombination von Kovarianzfunktionen dieses Problem lösen können. Daher ist die Untersuchung dieser Kombinations-Kovarianzfunktionen entscheidend für die Entwicklung robuster Modelle für scRNAseq-Daten. Hier erforschen wir die additive Kombination der häufig verwendeten Squared Exponential- (SQE) und Matérn 3/2-Kovarianzfunktionen und präsentieren eine neue Variante davon. Um die Unsicherheit bei der Schätzung der Modellhyperparameter zu quantifizieren, verwenden wir Bayessche Inferenz, um die Modelle aus den Daten zu lernen. Zur Untersuchung der Auswirkungen der vorgeschlagenen Kovarianzfunktionen haben wir eine umfangreiche Simulationsstudie durchgeführt, in der wir die Genauigkeit der Wiederherstellung der wahren Hyperparameter und die Vorhersageleistung bewerten. Darüber hinaus wurden zwei Fallstudien durchgeführt, um die Kovarianzfunktionen an echten scRNAseq-Datensätzen zu bewerten. Durch unsere Analyse bestätigen wir, dass die häufig verwendete SQE-Kovarianzfunktion nicht in der Lage ist komplexe Daten zu erfassen. Basierend auf den Fallstudien können wir jedoch nicht bestätigen, dass die additive Kombination von Kovarianzfunktionen besser geeignete Modelle für scRNAseq-Daten produziert. Angesichts der ähnlichen Leistung der Standard-Kovarianzfunktionen stellt sich die Frage, ob die Entwicklung von Kombinations-Kovarianzfunktionen tatsächlich notwendig ist, da sie die Modellkomplexität erhöht und nur ein begrenztes Verständnis des zugrunde liegenden biologischen Prozesses bietet.

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List of Abbreviations

ELPD	Expected log pointwise predictive density
GP	Gaussian process
HMC	Hamiltonian (or Hybrid) Markov chain
LOOCV	Leave-one-out cross-validation
LPD	Log pointwise predictive density
MOGP	Multi-output Gaussian process
scRNAseq	Single-cell RNA sequencing
SD	Standard deviation
SQE	Squared exponential

Notation and Symbols

Symbol	Explanation
$O(\cdot)$	Big-O notation for runtime and space complexity
$\text{cholesky}(K)$	Cholesky decomposition of the matrix K
$\text{Cov}[\cdot, \cdot]$	Covariance of two terms
$\mathbb{E}[\cdot]$	Expected value
$\text{Sd}(\cdot)$	Estimator function of the standard deviation
$K(t_i, t_j)$	Covariance function evaluated for time points t_i and t_j
Id	Identity matrix
$K_i \otimes K_j$	Kronecker product of matrix K_i and K_j
$\text{LKJ}(\cdot)$	Lewandowski-Kurowicka-Joe distribution
$\text{unif}(a, b)$	Uniform distribution ranging from a to b
α	GP marginal standard deviation
ρ	Length scale parameter of a covariance function
σ	Noise standard deviation
$\backslash$	Left matrix divide
$\sim$	Distributed according to

Chapter 1

Introduction

In recent years, single-cell RNA sequencing (scRNAseq) became the state-of-the-art method for investigating the diverseness of tissues and other biological samples on a single-cell resolution [Jovic et al., 2022]. The quantification of gene expression for thousands of genes in millions of single cells allows for various gene- and cell-level analyses [Rostom et al., 2017], making scRNAseq a valuable tool for many biologists and immunologists [Stuart and Satija, 2019]. It generally allows for the detection of novel and rare cell types [Jindal et al., 2018], and differential gene expression analyses [Gupta and Kuznicki, 2020]. Recent developments also allow the study of differentiation processes and the dynamics of biological processes through concepts like pseudotimes and RNA velocities [Haghverdi et al., 2016, Bergen et al., 2020]. Thus, further development of models for this type of data will allow us to better analyse it by giving deeper insights into biological processes, thereby bringing forth progress in many biological and medical fields.

1.1 Biological Background

Computational methods enable the estimation of pseudotimes, which is a concept used to temporally order the single cells along a certain trajectory representing a biological process [Trapnell et al., 2014, Haghverdi et al., 2016, Gupta et al., 2022]. By temporally ordering the cells also the dynamics of gene expression over the biological process can be portrayed and studied. This allows for analyses of branching developmental processes for example in cell

differentiation data and the identification of transient cell states as well as branching points and endpoints [Haghverdi et al., 2016, Gupta et al., 2022].

By utilising the information about the spliced and unspliced mRNA counts from a scRNAseq dataset an RNA velocity can be defined, describing the change of mRNA abundance over pseudotime [Bergen et al., 2020]. Mathematically, this RNA velocity can thus be considered the temporal derivative of the mRNA abundance over pseudotime. Due to the destructive nature of single-cell RNA sequencing, the resulting datasets are just snapshots of biological processes. Therefore they do not directly contain information about the underlying dynamics. With pseudotime and RNA velocity estimation methods, one can easily infer the dynamics and allow for a better description of this underlying biological process. However, due to the complex structure of biological processes and the inherent issues of scRNAseq data, these methods have limitations and do not always deliver accurate estimations of the pseudotime or RNA velocities [Gorin et al., 2022, Bergen et al., 2021]. Because of biological and many technical issues in the generation of these datasets, scRNAseq data suffers from a high dropout, a low sequencing depth and general noise in the resulting count tables. For further advancement in this field, the development of better-performing and more robust models is crucial.

1.2 Gaussian Processes

Gaussian processes (GPs) represent a versatile and powerful family of models that can be utilised to model scRNAseq data and tackle the aforementioned issues. They are non-parametric probabilistic models, which utilise a function f defined over an index set like the real numbers $\mathbb{R}$ to model observed data points. Because they can be thought of as a distribution over functions (see Figure 2.1), they model the observed data probabilistically. Gaussian processes are very flexible and find application in various scientific fields for modelling time series data, spatial data and general dynamics [Rasmussen and Williams, 2006, Duvenaud, 2014].

Unlike common regression models, which make many assumptions about the data that is being modelled, Gaussian processes relax these assumptions making them more versatile. Due to this, GPs can be used to model autocor-

related time-series data. Therefore, these relaxations are one of the primary motivations for using GP models for scRNAseq data, because it is by nature autocorrelated. Derivative GPs extend the base Gaussian processes and allow the simultaneous modelling of observed data and respective derivative information. Since RNA velocities can be seen as the temporal derivative of mRNA abundance, being able to utilise it for modelling scRNAseq data is another reason for GP models. Furthermore, as Gaussian processes are probabilistic methods, they inherently account for uncertainty and variability in the data, a crucial aspect for interpreting results correctly. Figure 2.1 (C) illustrates this quantified variance of the modelled underlying function of a GP fitted to data.

Despite their numerous advantages for modelling scRNAseq data, Gaussian processes also have some limitations. The largest limitation of Gaussian processes is their bad scalability of the runtime and space complexity. For a dataset with N data points, exact GP models have a runtime complexity of $O(N^3)$ and space complexity of $O(N^2)$ [Rasmussen and Williams, 2006]. In the case of multi-output GPs, which are used here and will be discussed later, these complexities increase to $O(N^3 D^3)$ for the runtime and to $O(N^2 D^2)$ for the space complexity with D being the number of modelled dimensions. Furthermore, to define a Gaussian process a covariance matrix is needed. This covariance matrix plays an important role in the performance of the GP models and usually has to be carefully picked manually [Duvenaud, 2014].

1.3 Covariance Functions

As mentioned, a covariance matrix is needed to define a Gaussian process. The detailed reason for this will be given in Chapter 2, to summarise, GPs are implemented using a multivariate normal distribution, which needs a covariance matrix. The covariance matrix plays an important role in a GP model because it encodes important properties of the function f that is used to model the presented data [Rasmussen and Williams, 2006, Duvenaud, 2014]. Some examples of properties that can be encoded are the periodicity, smoothness and stationary of the function f . Therefore, using the correct covariance function of the models plays an important role in the model's goodness of fit [Duvenaud, 2014].

Usually, the covariance matrices used for GP models are calculated with covariance functions. Two standard covariance functions further investigated in this thesis are the squared exponential (SQE) and the Matérn 3/2 covariance functions. GPs defined with the SQE covariance function will yield a highly smooth function f , as seen in Figure 2.1 (A), often considered too smooth for modelling real-world data [Stein, 1999]. The Matérn 3/2 covariance function, on the other hand, will yield a less smooth function, as seen in Figure 2.1 (B) [Matérn, 1960].

Combining multiple covariance functions results in the combination of their encoded properties, thereby giving rise to more complex properties [Duvenaud, 2014]. This property is often needed to effectively model data exhibiting intricate dynamics. Biological processes can be very complex and thus lead to complex dynamics of RNA expression. To model these complex dynamics effectively with Gaussian processes, appropriate covariance functions are needed. Therefore, the goal of this thesis is to investigate whether a combination of standard covariance functions will result in better Gaussian process models for scRNAseq data. Specifically, we will study the additive combinations of the SQE and the Matérn 3/2 covariance function in multiple study setups to obtain various conclusions about the possible gains of these proposed combinations.

1.4 Bayesian Inference

In this thesis, we utilise Bayesian inference as the method of choice to learn the appropriate hyperparameters of the models based on the observed data. In comparison to estimation methods regarding, for example, the maximum likelihood, Bayesian inference is an asymptotically unbiased inference method [Gelman et al., 2021]. The general concept is to use a sampling algorithm, like Markov chain Monte Carlo, to sample likely combinations of hyperparameters from the joint hyperparameter space. Further details of the method will be given in Chapter 2.

Here, we chose to use Bayesian inference to fit our proposed models, because of its many advantages. First, Bayesian inference methods are highly flexible and can be used with complex models such as the here-used Gaussian

processes. Second, it allows for the integration of prior knowledge about the model and the data, which is particularly important when dealing with limited data. Furthermore, Bayesian inference provides a direct way to quantify the uncertainty of the estimated hyperparameters by expressing them as probability distributions. Combined with the concept of confidence intervals, this allows us to make claims of the following form: *"the true parameter has a probability of 0.95 of being in the 95% confidence interval"*. The main disadvantage of Bayesian inference is the computational complexity of the commonly used Markov chain Monte Carlo sampling methods. However recent advancements in these sampling techniques have enabled the fitting of more complex models, including those presented in this thesis, within more appropriate computation times. [Hoffman and Gelman, 2014, Stan Development Team, 2023b].

1.5 Contributions

- We established a Gaussian process model for probabilistically modelling single-cell RNA sequencing data. We simultaneously integrate the measured gene expression, the estimated RNA velocities as the derivative of the gene expression and the estimated pseudotimes as the temporal ordering of the cells along the biological process.
- We propose two additive combinations of standard covariance functions and expand them for use in derivative GPs.
- With an extensive simulation study including a known ground truth data generating process, we show that the standard SQE covariance function is unable to model complex data well.
- Through a further case study, we show that the proposed additive combinations do not result in better models of scRNAseq data.

1.6 Structure of Thesis

In Chapter 2 we will first explain the proposed Gaussian process model in detail by explaining all modifications made and give reasons for why they

were made. Then, we will elaborate on the standard covariance functions used and propose the two additive combinations that will be investigated. Finally, an overview of the Bayesian framework used to learn the model's hyperparameters will be given. In Chapter 3 we will outline the simulation study and show how the models were compared based on their performance of the ground truth hyperparameter recovery. We will directly show and discuss the results to motivate the further performed analysis. In Chapter 4 we will explain the measures used to compare the models based on their predictive performance. Furthermore, we will use these measures to compare the models on some simulated datasets and discuss the results. After that in Chapter 5, we will use the same measures in a case study with two scRNAseq datasets. Finally, we will collect all conclusions from the separate analyses and give an overall conclusion with an extensive outlook for further investigation and research.

Chapter 2

Material and Methods

In this chapter, we will introduce the final Gaussian process model used in this thesis and provide reasons for all model decisions made. After that, we will introduce the investigated covariance functions and further explain their properties. Finally, we will state the Bayesian framework in which we fit our models to data and explain the measures used to validate the fits.

2.1 Gaussian Process

A Gaussian process is an indexed set of random variables $\{X_t\}_{t \in T}$ with the condition that for each finite subset of indices $t_1, \dots, t_m \in T$ the random variable $(X_{t_1}, \dots, X_{t_m})$ follows a multivariate normal distribution [MacKay, 2003]. Therefore, a Gaussian process is a joint distribution of infinitely many random variables and can be seen as a distribution of functions defined over T . An example of such a distribution can be seen in Figure 2.1 (A). For this thesis, the index set T is generally assumed to be the real numbers $\mathbb{R}$. Usually, GPs are used to model a finite dataset and, thus are represented by a multivariate normal distribution. Therefore, a Gaussian Process can be fully defined by a mean function $\mu(\cdot)$ and a covariance function $K(\cdot, \cdot)$ [Rasmussen and Williams, 2006]. The mean function $\mu(\cdot)$ defines a mean value for each index $t \in \mathbb{R}$ that is the mean value of functions drawn from a respective Gaussian process at point t . Here, we investigate zero-mean GPs and thus fix the mean function to zero. The covariance function, on the other hand, defines the covariance of two function values at points t_i

and t_j . Assuming that a random function f is distributed according to a Gaussian process with mean function $\mu(\cdot)$ and covariance function $K(\cdot, \cdot)$, noted as $f \sim \mathcal{GP}(\mu, K)$, then

$$\begin{aligned}\mathbb{E}[f(t)] &= \mu(t) \quad \forall t \in \mathbb{R}, \\ \text{Cov}[f(t_i), f(t_j)] &= K(t_i, t_j) \quad \forall t_i, t_j \in \mathbb{R}\end{aligned}$$

holds true [Rasmussen and Williams, 2006]. Only positive semidefinite functions generate positive semidefinite covariance matrices and thus are valid covariance functions [Rasmussen and Williams, 2006]. Because covariance functions define the similarity of two function values $f(t_i)$ and $f(t_j)$ they directly encode assumptions about f like the smoothness or periodicity.

Usually, we use the true function f represented by a GP to model a finite number of noisy observations y_i for each index t_i with $i \in \{1, \dots, N\}$ that is assumed to be centred around $f(t_i)$. Therefore, the noise ε_i is normally distributed with a zero mean and an unknown variance σ^2 , so that

$$y_i = f(t_i) + \varepsilon_i.$$

In GPs, we can directly incorporate this noise by adding σ^2 to the main diagonal of the covariance matrix used by the multivariate normal distribution [Rasmussen and Williams, 2006].

2.1.1 Derivative Gaussian Process

When Gaussian processes are subjected to linear operations, the result is again a Gaussian process [Solak et al., 2002, Rasmussen and Williams, 2006]. Because differentiation is a linear operator, the derivative of a Gaussian process is again another Gaussian process with a related mean and covariance function [Rasmussen and Williams, 2006]. Furthermore, when combining function points $f(t)$ and their respective derivatives $\frac{\partial f(t)}{\partial t}$, from now on denoted as f' , a joint Gaussian process can be defined over them. This allows for the simultaneous modelling of function values and their derivatives [Rasmussen and Williams, 2006, Solak et al., 2002, Rasmussen, 2003, Padidar et al., 2021]. An example of this joint model can be seen in Fig-

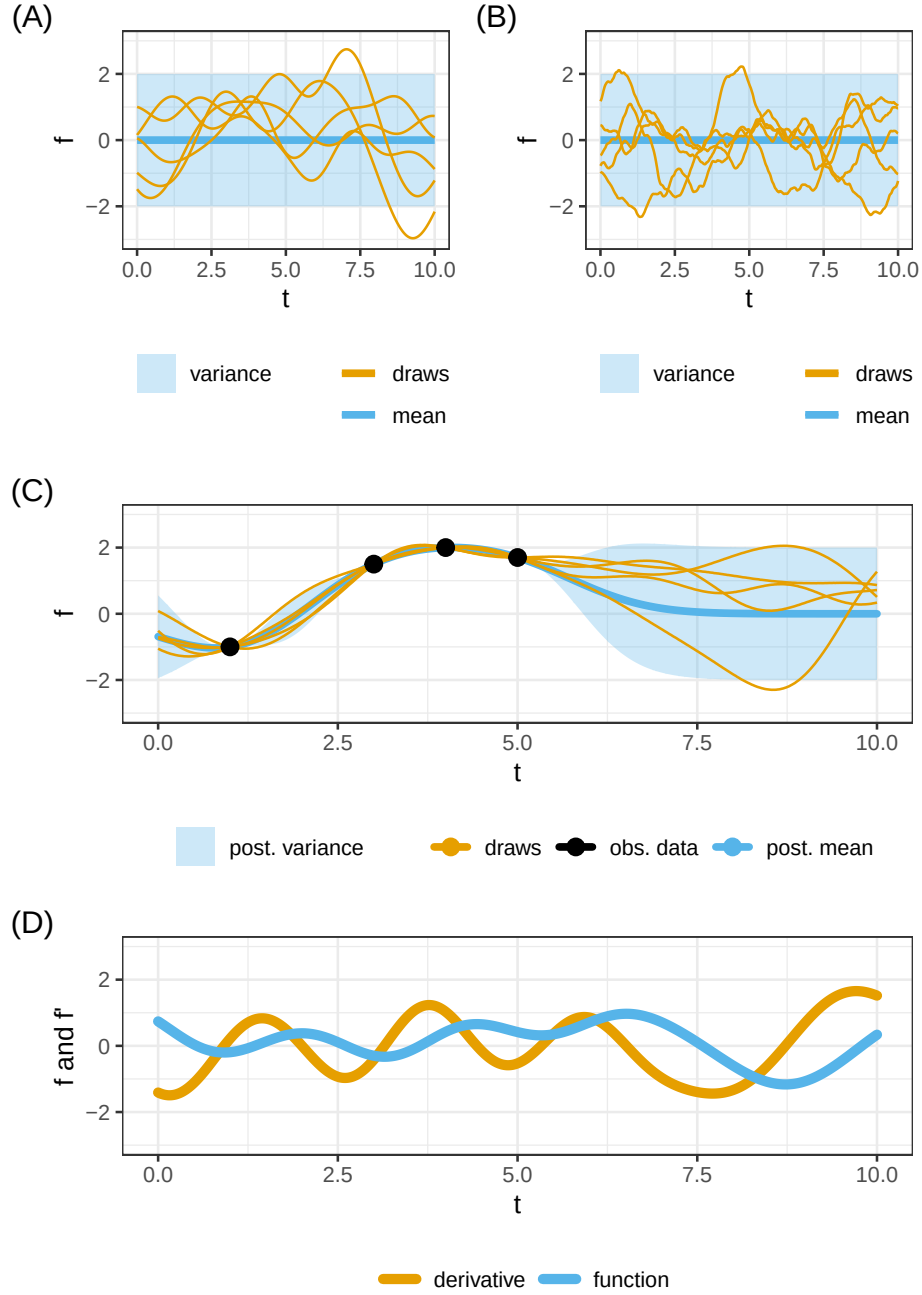


Figure 2.1: Examples of different Gaussian processes

Figures (A) and (B) show examples of zero-mean GPs with a GP marginal SD $\alpha = 1$ with the SQE covariance function (A) and the Matérn 3/2 covariance function (B). In Figure (C) a noise-free GP conditioned of data is portrayed. Figure (D) shows a sampled function from a derivative GP.

ure 2.1 (D).

Given a Gaussian process with a twice differentiable covariance function K , which defines the covariance of function values $f(t)$, the joint derivative Gaussian process can be constructed as follows:

$$\begin{pmatrix} f(t) \\ f'(t) \end{pmatrix} \sim \mathcal{GP} \left(0, \begin{pmatrix} K & K'_1 \\ K'_2 & K'' \end{pmatrix} \right). \quad (2.1)$$

The introduced sub-covariance matrices K'_1 , K'_2 and K'' are given by:

$$\text{Cov}[f(t_i), f'(t_j)] = \frac{\partial K(t_i, t_j)}{\partial t_j} =: K'_1 \quad (2.2)$$

$$\text{Cov}[f'(t_i), f(t_j)] = \frac{\partial K(t_i, t_j)}{\partial t_i} =: K'_2 \quad (2.3)$$

$$\text{Cov}[f'(t_i), f'(t_j)] = \frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} =: K'' \quad (2.4)$$

and define the proper covariances between all combinations of function and derivative values [Mukherjee et al., 2024]. A derivative Gaussian process can be defined like this, as long as the covariance function K is twice differentiable. The derivatives for all covariance functions investigated in this thesis and their derivation are shown in Appendix A.1.

To extend the notation of the observed data (y_i, t_i) for $i \in \{1, \dots, N\}$ to include the observed derivative information we denote y'_i as the observed value for the derivative $\frac{\partial f(t_i)}{\partial t_i}$. Therefore, the data modelled with derivative GPs is (y_i, y'_i, t_i) for $i \in \{1, \dots, N\}$. Similar to the handling of the noise for the normal GPs, here σ^2 is added to the main diagonal of all the submatrices shown in equation 2.1 [Mukherjee et al., 2024].

Adapted Parameters

The magnitudes of the function $f(t)$ and its derivative $\frac{\partial f(t)}{\partial t}$ can vary greatly, because of inherent properties of differentiation. Thus, utilising two standard deviation (SD) parameters of the GP marginal and the error is needed to properly model functions and their respective derivatives [Mukherjee et al., 2024]. Here, we denote the two GP marginal SD parameters as α and α' , and the two error SD parameters as σ and σ' . We utilise them similarly to

Mukherjee et al., 2024.

2.1.2 Multi-output Gaussian Process

GPs can be extended to define a distribution over a multi-dimensional function $f^* : \mathbb{R} \rightarrow \mathbb{R}^d$. These so-called multi-output Gaussian processes (MOGPs) can be constructed in many ways with varying assumptions in mind [Liu et al., 2018]. In this thesis, the linear model of coregionalisation will be utilised [Teh et al., 2005]. The general approach relies on modelling the function $f : \mathbb{R} \rightarrow \mathbb{R}^d$ as d independent GPs that are linearly combined [Liu et al., 2018]. Here, independent GPs with their own sets of hyperparameters will be used and linearly combined using a correlation matrix $\Omega \in \mathbb{R}^{d \times d}$ [Teh et al., 2005, Bonilla et al., 2007]. This way, the correlation structure of the separate dimensions is directly modelled. For an efficient implementation, we use the Cholesky factorisation of the correlation matrix Ω to apply the correlation between the independent outputs [Stan Development Team, 2023b, Mukherjee et al., 2024].

The MOGP has to be formulated with only one covariance matrix, to allow an efficient computation of its predictive performance for later analyses. The new all-in-one covariance matrix needs to account for the correlation of the different dimensions among themselves and the covariances of each dimension with its respective hyperparameters. Let $K^{(i)}$ be the covariance matrix of the i th latent GP and K the block diagonal matrix containing all $K^{(i)}$. Then the all-in-one covariance matrix is defined as

$$\tilde{K} = (\Omega \otimes \text{Id})K(\Omega \otimes \text{Id}) \quad (2.5)$$

with $\otimes$ being the Kronecker product and Id being the identity matrix [Teh et al., 2005]. The variables y_i and y'_i of the observed data points now denote d -dimensional vectors instead of single values as before.

2.1.3 Final Model

The scRNAseq data we want to model in this thesis contains expression values, respective RNA velocity estimates and pseudotimes for multiple cells and

genes. Therefore, the final Gaussian process model we use here is a composite of the derivative GP with adapted parameters and the correlated multi-output GP. A derivative GP is needed to incorporate the RNA velocities as an estimate of the change of gene expression over the pseudotime. Since the RNA velocities are just an estimate of the actual derivative and can change in their scale, we need the adapted parameters of the derivative GP to properly account for the scale difference. ScRNAseq data incorporates numerous gene expression values for each cell, thus a MOGP is needed to model multiple genes simultaneously. Since the gene expression and their described dynamics can vary greatly for different genes, each gene must be modelled with its respective set of hyperparameters. Furthermore, it is known that genes are correlated, thus modelling the correlation structure within the MOGP is also important. The only difference in the models we will compare in this thesis is the respective covariance function used.

2.1.4 Inference

One convenient feature of GPs is the existence of an analytical solution of the posterior GP [Rasmussen and Williams, 2006]. The posterior GP is a predictive Gaussian process that is conditioned on observed data points. An example of a posterior GP can be seen in Figure 2.1 (C). The posterior GP is represented by a posterior mean vector μ_{post} and a posterior covariance matrix K_{post} . By utilising the Cholesky factorisation they can be efficiently computed [Rasmussen and Williams, 2006]. Given the covariance matrix of the observed data points K and the covariance matrix of the observed with the to-be-inferred data points K_* the posterior mean is calculated as

$$\begin{aligned} L &:= \text{cholesky}(K + \sigma^2 \text{Id}) \\ L_* &:= L^\top \setminus (L \setminus y) \\ \mu_{\text{post}} &:= K_*^\top L_* \end{aligned}$$

with $\backslash$ being a left matrix divide [Rasmussen and Williams, 2006]. The posterior covariance matrix is calculated as

$$\begin{aligned} v &:= L \backslash K_* \\ K_{\text{post}} &:= K_{**} - v^\top v \end{aligned}$$

with K_{**} being the covariance matrix of the to-be-inferred data points [Rasmussen and Williams, 2006]. With the covariance function formulations in equations 2.2 to 2.4 and 2.5 this makes it also applicable for the full model considered in this thesis.

2.2 Covariance Functions

As already stated in 2.1, to define a GP a positive semidefinite covariance function has to be given [Rasmussen and Williams, 2006]. In the following, the covariance functions used in this thesis are introduced and their properties explained. We only investigated stationary covariance functions, which depend solely on the difference between data points $t_i - t_j$ and are thus invariant to translations of these [Rasmussen and Williams, 2006].

2.2.1 Squared Exponential

The squared exponential (SQE) covariance function, also called radial-basis function kernel or Gaussian kernel [Duvenaud, 2014], is one of the most-used covariance functions in GP applications [Rasmussen and Williams, 2006]. The exact formulation and the derivatives that are needed for the definition of a derivative GP are given by

$$\begin{aligned} K(t_i, t_j) &= \alpha^2 \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right), \\ \frac{\partial K(t_i, t_j)}{\partial t_j} &= \alpha^2 \frac{t_i - t_j}{\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right), \\ \frac{\partial K(t_i, t_j)}{\partial t_i} &= -\alpha^2 \frac{t_i - t_j}{\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \text{ and} \\ \frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} &= \frac{\alpha}{\rho^4} \cdot (\rho^2 - (t_i - t_j)^2) \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right). \end{aligned}$$

The respective derivations of the derivatives can be found in Appendix A.1.1. The SQE covariance function has two parameters, the GP marginal SD α , and the length scale ρ . Given $f \sim \mathcal{GP}(\mu, K)$, the GP marginal SD α describes how strong f can vary from the mean function μ . GPs with low α will produce functions close to μ whereas a high value of α will produce functions that can vary a lot from μ . The length scale ρ describes the smoothness of the function f . A high length scale value corresponds to functions that are smoother than functions drawn from a GP with a lower length scale.

The SQE covariance function is infinitely many times differentiable and therefore defines a distribution of very smooth functions, which again are infinitely many times differentiable. When modelling real-world data this extreme smoothness is often unrealistic [Rasmussen and Williams, 2006, Stein, 1999].

2.2.2 Matérn 3/2

Compared to the SQE covariance function the Matérn 3/2 covariance function generates less smooth functions as can be seen in Figure 2.1 (A) and (B). It belongs to the Matérn class covariance functions [Stein, 1999, Matérn, 1960] and is the least smooth one apart from the Matérn 1/2, which is very rough [Rasmussen and Williams, 2006]. This lower smoothness allows GPs defined with it to model more complex data, which is the main reason why we included the Matérn 3/2 covariance function in this thesis. The equations needed for the final model are given by

$$\begin{aligned}
K(t_i, t_j) &= \alpha^2 \cdot \left(1 + \frac{\sqrt{3} \cdot (t_i - t_j)}{\rho} \right) \cdot \exp \left(-\frac{\sqrt{3} \cdot (t_i - t_j)}{\rho} \right), \\
\frac{\partial K(t_i, t_j)}{\partial t_j} &= \alpha^2 \cdot \left(\frac{3(t_i - t_j)}{\rho^2} \right) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)}{\rho} \right), \\
\frac{\partial K(t_i, t_j)}{\partial t_i} &= -\alpha^2 \cdot \left(\frac{3(t_i - t_j)}{\rho^2} \right) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)}{\rho} \right) \text{ and} \\
\frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} &= \alpha^2 \cdot \left(\frac{3}{\rho^2} \right) \cdot \left(1 - \frac{\sqrt{3}(t_i - t_j)}{\rho} \right) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)}{\rho} \right).
\end{aligned}$$

Again, the respective derivations can be found in Appendix A.1.2. The Matérn 3/2 covariance function has the same parameters as the SQE. Their influence on the functions drawn from a GP is equal to the ones in the SQE covariance function and is explained in 2.2.1.

The Matérn 3/2 covariance function is twice differentiable [Rasmussen and Williams, 2006] and gives rise to less smooth functions when used as the covariance function of a GP. This reduced smoothness might be more realistic for modelling real-world data [Stein, 1999] and is the primary reason for its investigation in this thesis.

2.2.3 Combination of Covariance Functions

Each type of covariance function encodes certain properties of the functions drawn out of a Gaussian process. By combining multiple covariance functions, also their encoded properties can be combined. Both addition and multiplication of covariance functions result again in valid covariance functions that are positive semidefinite [Rasmussen and Williams, 2006]. As long as the covariance functions being combined are twice differentiable, the combination of them will still be twice differentiable. This becomes evident from the derivatives of the combined covariance functions that can be seen in Appendix A.1.3.

In this thesis, we primarily investigate the additive combination and additionally propose a variant with fewer degrees of freedom. Although a multiplicative combination of covariance functions can bring forth more interesting properties, it was kept out of the scope of this thesis due to a more complex derivative structure as seen in Appendix A.1.3. The additive combination of two covariance functions K_1 and K_2 is defined as

$$K_{\text{add}} := K_1 + K_2.$$

Here, both covariance functions have their own GP marginal SD parameter α and α' parameters, resulting in 4 hyperparameters ($\alpha_1, \alpha_2, \alpha'_1, \alpha'_2$) per modelled dimension that account for the GP marginal SD. The modification that we propose in the thesis is the mixing of two covariance functions

$$K_{\text{mix}} := \alpha_{\text{gen}}(m \cdot K_1 + (1 - m) \cdot K_2)$$

with $m \in [0, 1]$ being the mixing parameter. Now, both covariance functions have fixed hyperparameters $\alpha = \alpha' = 1$ and the general GP marginal SD is defined by the α_{gen} . This way, 3 hyperparameters $(\alpha_{\text{gen}}, \alpha'_{\text{gen}}, m)$ instead of 4 control the GP marginal SD. The motivation for this restriction is that we assume the ratios between α and α' for one of the two covariance functions in the additive combination to be the same as for the second covariance function.

2.3 Bayesian Inference

Bayesian inference is a statistical method for fitting a probabilistic model to observed data y . In our case, all hyperparameters of the models, from now on denoted as θ , are fitted to the observed function values y and their estimated derivative values y' . The hyperparameters θ include the GP marginal SDs, length scales and noise SD of each dimension, and the correlation matrix of the dimensions. The essence of Bayesian inference is to calculate the distribution of parameters given the observed data using Bayes' theorem [Gelman et al., 2021]. This distribution is called the posterior distribution and is given by

$$p(\theta|y, y') = \frac{p(\theta, y, y')}{p(y, y')} \propto p(y, y'|\theta) \cdot p(\theta).$$

This posterior distribution encapsulates our updated beliefs about the parameters after taking the observed data into account. The prior distribution $p(\theta)$ must be given before fitting the models and is further explained in 2.3.1. Generally, the true posterior distribution is asymptotically approximated with Bayesian sampling algorithms like Markov chain Monte Carlo (MCMC) methods producing posterior samples of θ [McElreath, 2020]. This has to be done because direct sampling from the posterior distribution is not possible due to its complexity and high dimensionality. Here, we used the Hamiltonian Monte Carlo, or Hybrid Monte Carlo, algorithm (HMC) [Duane et al., 1987] because of its highly efficient implementation of MCMC sampling [McElreath, 2020]. This algorithm produces less autocorrelated samples and thus is more efficient compared to other methods [Gelman et al., 2021].

2.3.1 Used Settings and Priors

We implemented all models with the statistical programming language Stan [Stan Development Team, 2023b] and its R interface [Stan Development Team, 2023a]. We used one single chain to generate 3000 samples from which the first 1000 were omitted as warm-up samples. The sampling was performed with only one chain to reduce the computation times. For the subsequent analyses, the same sampling settings and priors were used.

We chose to use weakly informative priors and normalised the pseudotime ranges of all used datasets to be able to use the same prior settings for all analyses. We used the inverse-gamma distribution with a fixed shape parameter of 5 for the length scale prior. The scale parameter β was chosen so that the resulting mean of the distribution is equal to the mean distance of all observed time points t_i . For the combined covariance function models, this approach led to non-convergence, thus we used a fixed scale parameter of 6 for these models. As priors for the SD of the GP marginal α, α' and noise σ, σ' we picked $\mathcal{N}(0, \text{Sd}(y))$, which is informed on the standard deviation $\text{Sd}(y)$ of the data y used for fitting. For the correlation matrix Ω we used the LKJ(3) distribution, which is a common prior for correlation matrices [Lewandowski et al., 2009].

2.3.2 Evaluation Metrics of Convergence

Since Bayesian sampling is not a deterministic method, the produced samples always vary and might be unreliable. Because of this, it is important to use evaluation metrics, which allow us to identify certain issues in the sampling process. In this thesis, we used the following two measures to quantify the goodness of the computed samples.

Potential Scale Reduction Statistic $\hat{R}$

Markov chain Monte Carlo methods do not sample immediately from the true posterior distribution. Often the chains are initialised at random positions and only reach the posterior distribution after an unknown number of samples. The general convention in Bayesian inference is to disregard the first n burn-in samples and only work with the remaining ones. Often, considering the first

1000 samples as a burn-in and removing them is sufficient. The motivation of the $\hat{R}$ statistic is to quantitatively measure this convergence on the posterior distribution [Gelman and Rubin, 1992].

The originally proposed $\hat{R}$ statistic fails when chains have different variances or the posterior distribution has heavy tails. For these reasons, Stan uses the Split- $\hat{R}$. Here, the chains are split up in two, before doing the rest of the calculations. This allows for detecting within-chain non-stationarity, resulting in a better convergence statistic.

The $\hat{R}$ and the split- $\hat{R}$ statistics can assume values ranging from 1 to ∞ , where smaller values mean a better convergence. For the original $\hat{R}$ a value of 1.1 was considered to indicate a good convergence [Gelman and Rubin, 1992]. Later, it was shown that this cut-off value was too soft and convergence can not be guaranteed. To securely indicate a convergence of the sampling process a value of 1.01 is needed [Vehtari et al., 2021]. In this thesis, we relax this criterion not only due to the complex model structure and low number of training data. The main reason is that for the simulation study, the ground truth parameters and the data-generating process were known and used for comparison. The terms $\hat{R}$ and split- $\hat{R}$ are used ambiguously in recent literature, here from now on we mean the split- $\hat{R}$.

Effective Sample Size S_{eff}

Samples generated by Bayesian sampling often show autocorrelation, which refers to the correlation between samples within the same chain separated by n steps. Even if HMC produces fewer autocorrelated samples, some autocorrelation is still present. A high autocorrelation indicates less independence between the values, which means that the posterior distribution of the samples is less accurate [Kruschke, 2015]. The effective sample size S_{eff} is a quantitative measure of this autocorrelation. Stan computes the effective sample size by combining the within-chain autocorrelation and cross-chain information from $\hat{R}$ [Vehtari et al., 2021]. In reality, Stan computes two types of the effective sample size. The bulk- S_{eff} is the general effective sample size measure described above, whereas the tail- S_{eff} measures sampling efficiency for the distribution tails [Vehtari et al., 2021].

To interpret the later produced results, larger values of bulk- S_{eff} and

tail- S_{eff} indicate a better approximation of the posterior distributions given a low $\hat{R}$ value. To have reliable estimates of the chain variance and autocorrelation it is recommended to have an average bulk- S_{eff} of 100 per chain [Vehtari et al., 2021].

Chapter 3

Simulation Study

3.1 Motivation and Setup

We performed a simulation study to evaluate the capabilities of true ground-truth hyperparameter recovery and thus how capable the models are in modelling the data. A simulated dataset with a known ground truth data-generating process is needed to allow a quantitative evaluation of the parameter recovery. Furthermore, such a simulation study gave us insight into the computation times of the models and how well they converge on the posterior distribution using the convergence measures.

In this thesis, we compared four derivative multi-output Gaussian processes with different covariance functions. The simple SQE covariance function was used as the baseline method. The Matérn 3/2 covariance function was picked as a slightly more complex method. The two other methods we propose are the addition and mixing of the two covariance functions mentioned. This way we can evaluate whether a combination of covariance functions is a sensible choice for modelling scRNAseq data.

To create the simulated datasets for the study, we randomly sampled hyperparameters from predefined distributions. The distributions from which the hyperparameters were sampled and other set values are stated in Table 3.1. We added a fixed scale factor of 3 to the standard deviation of the GP marginal α and the noise σ to include the scale difference between observed function values y and the observed derivative values y' . Then, we used these to sample a random set of data points from a generative Gaussian process.

Table 3.1: Hyperparameter sampling for simulated dataset

The hyperparameters were either randomly sampled from a distribution or set to a fixed value.

Hyper-parameter	Sample distribution	Fixed value & dependent parameter
α_{exp}	unif(3, 6)	-
α_{vel}	-	$\alpha_{\text{exp}}/3$
mix	unif(0.25, 0.75)	-
ρ_{SQE}	unif(0.5, 2)	-
ρ_{Matern}	unif(0.5, 2)	-
σ_{exp}	unif(0.5, 1.5)	-
σ_{vel}	-	$\sigma_{\text{exp}}/3$
correlation between dimensions	-	0.5

For this, we used the final model, described in 2.1.3, with 5 dimensions and the mixed covariance function. To keep the computation time of the whole study manageable, we generated 50 datasets each containing 20 data points. The data points were sampled equidistantly to range from 0 to 10 uniformly. Each data point resembles a cell, thus containing 5 expression values and their respective derivatives.

Before any comparison between the models was performed, some of the 50 datasets were removed from the analysis, because they led to non-convergent models. We used the mean $\hat{R}$ of all the relevant parameters (see Table 3.1; excluding correlation between dimensions) and the cut-off value 1.1 (see 2.3.2) to detect non-convergence. To evaluate the ground-truth parameter recovery we chose to calculate the mean absolute error

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n (\theta - \hat{\theta}).$$

With θ being the ground truth value of a parameter and $\hat{\theta}$ being its respective estimated value. The mean was computed over the posterior draws for each hyperparameter, model and dataset. We use the MAE for this analysis because it is an easy-to-interpret measure of the absolute error.

3.2 Results and Discussion

As discussed, we left out datasets that led to models with a mean $\hat{R}$ over all relevant hyperparameters greater than 1.1, thus indicating non-convergence. This was the case for 12 datasets, leading to 38 used for further analysis. Due to the large runtime of the models and the time constraints of this thesis, we were not able to simulate more datasets and fit the models on them.

3.2.1 Convergence of Bayesian Inference

In Figure 3.1 the convergence metrics and other statistics of the Bayesian inference are summarised as distributions over the individual 38 trials per model. The main proportion of $\hat{R}$ values for all models is close to 1, indicating good convergence on the posterior distribution. Few outlying $\hat{R}$ values were expected due to the complex structure of the multi-output derivative GP model. The lowest mean of the $\hat{R}$ values is visible for the SQE model.

For the effective sample sizes S_{eff} , the simplest model with the single SQE covariance function yields the highest values. Between the two combination covariance models the mixing one yields on average a slightly better bulk- S_{eff} and tail- S_{eff} than the additive one. The additive covariance function shows on average the lowest bulk- S_{eff} and tail- S_{eff} out of the compared models. This indicates that the additive covariance function generated the most autocorrelated posterior samples. The effective sample sizes fulfil the recommended value of 100 per chain with few exceptions in the more complex models.

In terms of runtimes, only minor differences between the models can be seen. The MCMC sampling for the SQE model is the fastest, followed by the Matérn 3/2 covariance function, the additive and the mixing combination. The presence of large outliers for the combination models suggests issues such as stuck chains for certain datasets.

3.2.2 Parameter Recovery

As seen in Figure 3.2, not all hyperparameters of each model were included in the comparison to the ground truth. The various models differ in their hyperparameters, therefore a comparison to the ground truth data, which was

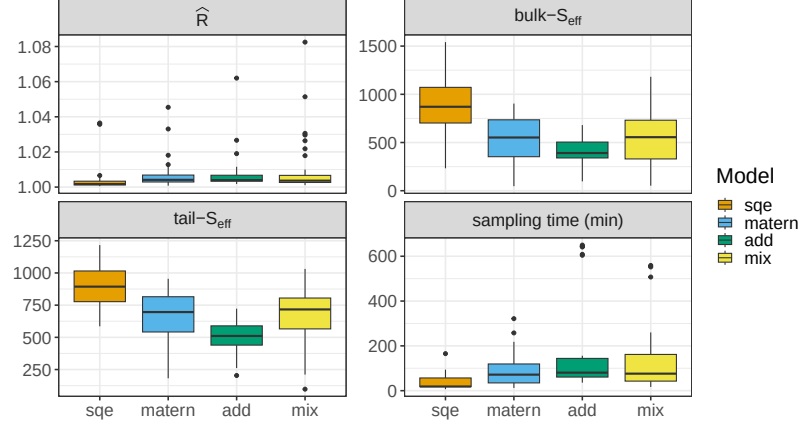


Figure 3.1: Convergence metrics and runtime for the models and simulated datasets

Not all 50 simulated datasets were considered, only the ones where the mean $\hat{R}$ of all relevant parameters was 1.1 or lower. Values for $\hat{R}$, bulk- S_{eff} and tail- S_{eff} shown here are the mean over all important parameters per trial and model.

generated with a mixing combination of two covariance functions, was not always sensible. The simple SQE and Matérn 3/2 models have only one covariance function, thus we did not compare them to either of the two ground truth lengthscales nor the mixing parameter. The goodness of ground truth hyperparameter recovery is measured with the MAE, hence lower values indicate a better recovery.

The GP marginal SDs α and α' have the lowest values for the mixing model and are thus best recovered by it (see Figure 3.2). The worst recovery is with the additive model because here the MAE values are overall higher than for the other models. Both baseline models perform equally well based on the MAE values and recover the ground truth hyperparameters worse than the mixing model but better than the additive one. Overall, the MAE values for α' are much smaller than for α , which was expected because of the scale difference between the two hyperparameters.

For the noise SDs σ and σ' the lowest MAE values and thus the best ground truth recoveries are visible for the additive model, closely followed by the mixing and the Matérn 3/2 one. All models have similar MAE distributions, except the SQE one, which has substantially larger values. This can mainly be explained by the fact that the SQE model is considered too smooth for

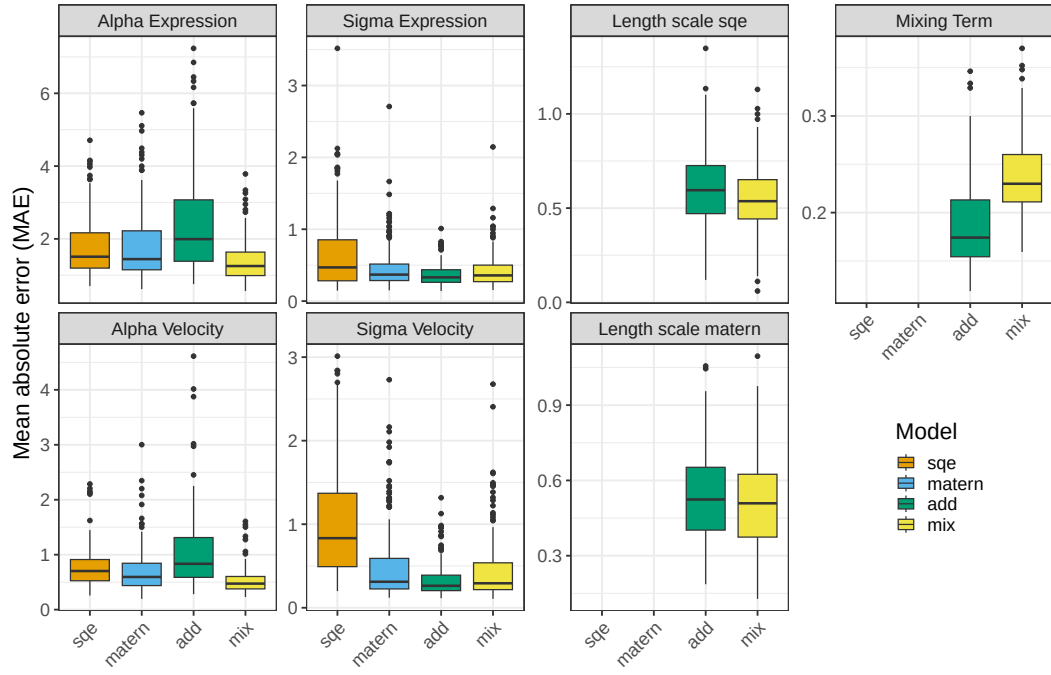


Figure 3.2: Mean absolute error distributions of parameters to the ground truth

Values for the different dimensions were grouped. The GP marginal SD α and the noise SD σ were present in all models and could be compared directly. For the two ground truth length scales a sensible comparison was only given for the additive and mixing covariance function models. The mixing parameter for the additive covariance function was derived by computing the ratio between the α of each single covariance function.

modelling complex data [Stein, 1999, Rasmussen and Williams, 2006]. Consequently, much of the data’s variance has to be explained by the noise, which, in turn, increases the error to the ground truth value. In Appendix A.2 we investigated this by directly comparing the σ , σ' and length scale distributions of the ground truth values with the mean of the posterior distribution of each trial. Figure A.1 clearly shows that the SQE model greatly overestimates the two ground truth length scales, indicating that this model is too smooth. Therefore, a larger portion of the variance in the data needs to be explained by the noise SD σ and σ' , which is confirmed in Figure A.2. Here we can see that the estimates for the SQE model are greatly larger than for the other models.

For the individual length scale and mixing hyperparameters, it made sense to compare only the additive and mixing models to the ground truth. From Figure 3.2 we can see that the mixing model can recover both length scales slightly better than the additive one. The additive model best recovered the mixing parameter of the simulated data, although the ground truth data-generating process was a GP with the mixing covariance function. The mixing in the additive model was computed after the fitting as the ratio of the single GP marginal SDs α_1 and α_2 . This was also done for the derivative GP marginal SDs α' to generate the mean ratio, which was used to calculate the MAE.

3.3 Conclusion

The simulation study was conducted to evaluate the overall performance of the models to fit data where ground truth for hyperparameters is present. To evaluate the hyperparameter recovery quantitatively, we used a generative Gaussian process with the mixing covariance function to generate data with known ground truth hyperparameters. We chose one of the complex covariance functions to have simulated data with complex properties. However, due to the inherent differences in the hyperparameters of the proposed models, a complete comparison of all models was not possible.

Nonetheless, these results enable us to draw certain conclusions about the models. From the distributions of convergence statistics in Figure 3.1 we can see that, under equal conditions, the simpler baseline models show a better convergence on the posterior distribution than the combination covariance

function models. This is reflected by the lower values of $\hat{R}$, which indicate a better convergence, and higher values of effective sample sizes S_{eff} signifying a better approximation of the true posterior distribution. Furthermore, the lower sampling times indicate that these models are also faster.

Moreover, we could confirm the widely accepted fact, that the SQE covariance function is often too smooth to model real-world data [Stein, 1999, Rasmussen and Williams, 2006]. From the parameter recovery analysis and the detailed investigation of the σ , σ' and length scale hyperparameters, it was observed that the SQE covariance function results in models that are excessively smooth and attribute much of the data's variance to noise. With this, it can be argued that the SQE covariance function is most likely not well-suited for modelling complex scRNAseq data.

A full comparison based on all important hyperparameters was only meaningful for the additive and mixing covariance functions because they share all hyperparameters. Generally, based on this simulation study no distinct preference for either the simple or the complex covariance functions can be determined. Therefore, we decided to extend the comparison of the individual models by further looking at their predictive capabilities.

Chapter 4

Leave-One-Out Cross-Validation

4.1 Motivation and Setup

In machine learning, leave-one-out cross-validation (LOOCV) is a commonly used technique to estimate prediction accuracy when a dedicated testing dataset is unavailable. LOOCV involves iteratively fitting the model on all but one data point. This left-out data point is then used to evaluate the model based on a performance measure. Due to the repeated re-fitting of the model and the extensive computation times associated with GPs, this approach is only feasible for small datasets.

The comparison of the models only based on the simulation study is restricted due to a large mismatch of the hyperparameters utilised by each model. Therefore, we used LOOCV to further compare the four models based on their predictive accuracies. For this, some of the datasets from the simulation study were used to perform multiple LOOCVs. Not all datasets were used because of the time constraints of this thesis. From all datasets generated for the simulation study, we used only the 5 with the lowest mean $\hat{R}$ across all models. These showed the best convergence and thus most reliable posterior distributions.

To compare the models, we used the expected log pointwise predictive density (ELPD), a measure of predictive accuracy [Vehtari et al., 2017]. In contrast to measures such as the root mean square error, this method considers the entire predictive distribution and is, therefore, preferred [Bürkner et al., 2021]. Since we are using Bayesian LOOCV, the metric we utilise is the LOOCV es-

timate of the true ELPD

$$\text{elpd}_{\text{loo}} = \sum_{i=1}^n \log p(y_i | y_{-i}).$$

This is the sum of all pointwise log predictive densities (LPD) with y_i being the left-out data point and y_{-i} the remaining data [Vehtari et al., 2017]. Generally, computing the log predictive density $\log p(y_i | y_{-i})$ is a hard task and thus can be approximated as

$$\log p(y_i | y_{-i}) \approx \frac{1}{S} \sum_{s=1}^S \log p(y_i | y_{-i}, \theta_{-i}^{(s)})$$

where S is the number of posterior draws and $\theta_{-i}^{(s)}$ the parameter values of the draw s from the posterior distribution [Bürkner et al., 2021]. Since Gaussian processes are modelled as multivariate normal distributions we, can directly calculate the log predictive density as

$$\log p(y_i | y_{-i}, \theta_{-i}^{(s)}) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \bar{c}_{ii} - \frac{1}{2} \frac{g_i^2}{\bar{c}_{ii}} \quad (4.1)$$

with

$$\begin{aligned} g_i &= [(K + \sigma^2 \text{Id})^{-1} y]_i \\ \bar{c}_{ii} &= [(K + \sigma^2 \text{Id})^{-1}]_{ii} \end{aligned}$$

where K is the respective covariance matrix of the model [Vehtari et al., 2016, Sundararajan and Keerthi, 1999]. Since one data point contains absolute and derivative values for multiple dimensions, equation 4.1 is computed individually and then summed up. In the end, we can compare the models based on their predictive capacities by comparing their elpd_{loo} values and the respective LPD distributions.

4.2 Results and Discussion

As mentioned, due to the re-fitting of the models and high computation times of GPs we only performed a LOOCV for 5 datasets from the simulation study.

Table 4.1: ELPD of models for five simulated datasets.

The best-performing model per dataset with the highest ELPD is shown in **bold**.

	dataset 13	dataset 14	dataset 19	dataset 23	dataset 47
sqe	-24761.4	-3867.6	-138374.5	-13294.1	-60406.1
matern	-1117.1	-967.1	-14828.1	-1666.1	-1697.7
add	-2790.8	-6684.9	-164123	-3356.7	-26163.6
mix	-3094.2	-2645.4	-65201.6	-4400.4	-69622.6

The resulting ELPD values for each model and dataset are given in Table 4.1, where the best-performing model is highlighted in bold. Furthermore, the whole distributions of the single LPD values per LOOCV-fold are shown in Figure 4.1. Since both the ELPD and the LPD are log-likelihoods, larger values indicate a better predictive capability.

In Table 4.1 we see that the Matérn 3/2 model has the highest ELPD values for each of the 5 datasets. Naturally, this effect is also visible in Figure 4.1, because the ELPD is the sum of all individual LPD values. From this, we can conclude that the Matérn 3/2 model has the best predictive performance on these specific simulated datasets with 20 data points. A likely explanation for this is that a GP with the Matérn 3/2 covariance function can inherently model more complex behaviour, as it exhibits less smoothness [Stein, 1999, Rasmussen and Williams, 2006].

Because the data was generated by a GP with the mixing covariance function, we expected the mixing and closely related additive model to perform better than the Matérn 3/2 one. The worse predictive performance, as indicated by the LPD values in Figure 4.1 can be explained by the higher complexity of the combination covariance function models. To fit more complex models, large training datasets are needed and we fitted the GPs only with 19 data points due to the left-out data point. Due to the time constraints of this thesis, we did not validate this probable explanation by repeating the LOOCV with more data points.

The SQE model performed, except for datasets 13 and 23, equally well as the combination covariance function models. Furthermore, the general ELPD values between the datasets vary substantially. To further investigate both discoveries, again a LOOCV with a larger dataset is needed.

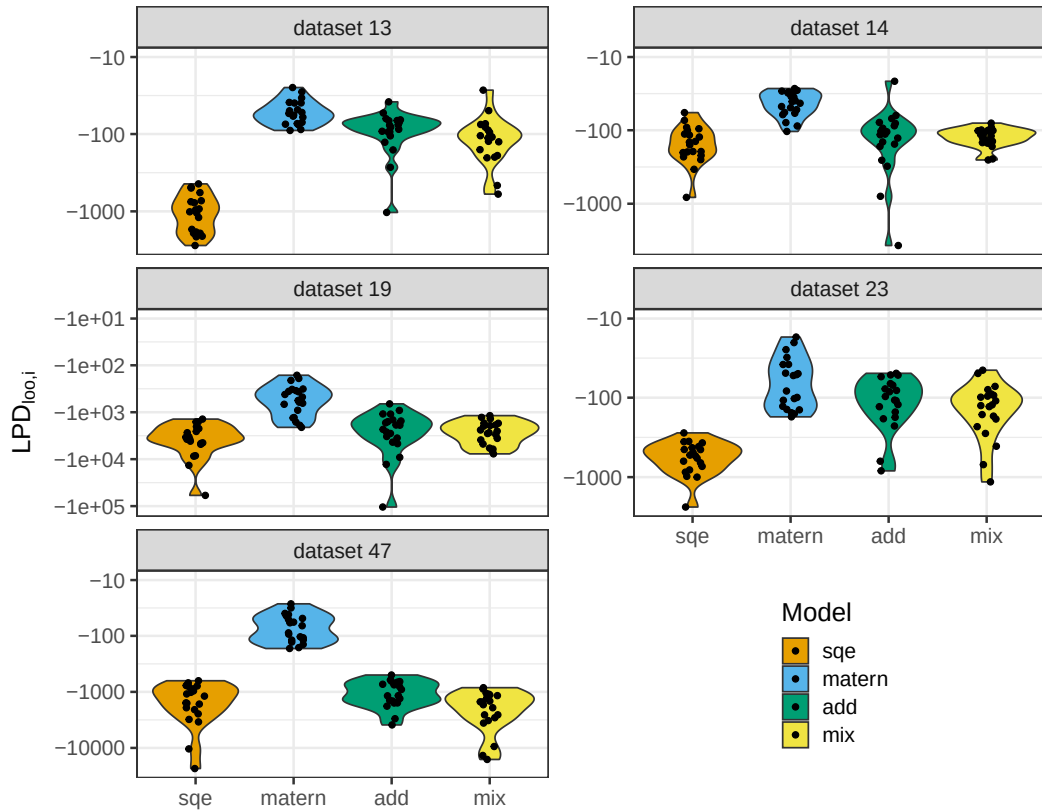


Figure 4.1: Log predictive density distributions for all models and five simulated datasets

In total a leave-one-out cross-validation with 20 data points was performed that resulted in the 20 shown LPD values. The violin plots summarise the distributions.

4.3 Conclusion

To conclude, we discovered that the Matérn 3/2 covariance function can model complex data well enough and performs better than the combination covariance functions for the simulated datasets. To make more concise conclusions, we would need to repeat the LOOCV with more data points, which was infeasible for us, due to time constraints. To see whether the same results considering the predictive capabilities of the models will be retained when fitted on actual scRNAseq data, we repeated the LOOCV study with some example datasets.

Chapter 5

Case Study

5.1 Data Processing

Until now we compared the models only on simulated datasets. We performed a case study to extend this assessment of the proposed models to real scRNAseq data and draw conclusions from it. For this purpose, we performed the LOOCV analysis with two example scRNAseq datasets. The datasets we used for this case study contain only simple biological processes that do not include the branching of cells into multiple sub-cell types. Furthermore, we decided to use openly available and already pre-processed datasets that were shown to represent specific known biological processes. The first dataset contains cells in the context of the cell cycle [Mahdessian et al., 2021] and the second, cells in response to neuronal activation [Qiu et al., 2020].

The datasets only contained values for the spliced and unspliced gene expression, thus we still needed to calculate the pseudotime and the RNA velocity. We estimated the RNA velocity with the widely used scvelo software [Bergen et al., 2020]. Instead of using the pseudotime generated by scvelo, we used Cytopath [Gupta et al., 2022] to estimate a different pseudotime shown to be biologically relevant. The datasets were processed as described by Gupta et al., 2022 using the respective openly available notebooks [Gupta et al., 2022].¹

After the RNA velocity and pseudotime were estimated, a subsampling of

¹The code is available at: <https://github.com/aron0093/cytopath-notebooks>

genes and cells was performed to ensure feasible runtimes of the individual LOOCVs. The cells were subsampled in a stratified manner to ensure a balanced distribution of data points across the whole pseudotime including all cell cycle stages. Since many genes had a high dropout or no significant gene expression change over the pseudotime, we manually chose 5 genes for each dataset that did not show these issues. For the cell cycle dataset, we chose the genes *CCNA2*, *CCNB1*, *BIRC5*, *TOP2A* and *RRM2*; for the neuronal activation dataset *HMGCS1*, *UBC*, *ATF4*, *HMGCR* and *CYP51* were chosen. As described, we scaled the pseudotime to $[0, 10]$ to standardise all datasets.

5.2 Results and Discussion

Due to the time constraint of this thesis, we initially only performed the LOOCV for each dataset with 20 data points. The resulting LPD distribution for each dataset can be seen in Figure 5.1. No significant differences in the LPD distributions between the proposed models are evident. Therefore, we cannot conclude which model performs better based on these case studies but can observe no gain in predictive performance from combining covariance functions. A possible reason for the similar LPD distributions is that the data might be too simple, thus able to be modelled sufficiently by all proposed GP models. To further validate this reason, an elaborate investigation of the model fits to the respective data is needed. Unfortunately, we were not able to do this due to the time constraint of this thesis.

As discussed in Chapter 4, due to the larger complexity of the combination covariance function models they may require more data points for accurately modelling the data. To validate whether a larger dataset will allow the models with combination covariance functions to model the data better than the simple ones, we repeated the LOOCV for the cell cycle dataset with 40 data points. As seen in Figure 5.1 this does not bring forth a difference between the LPD distributions of the models. This reinforces the claim that the additive models do not provide better models for scRNAseq data when compared to the standard covariance functions.

Generally, we can see an increase in the LPD values when fitted with 40 instead of 20 data points. This effect was expected because the models have

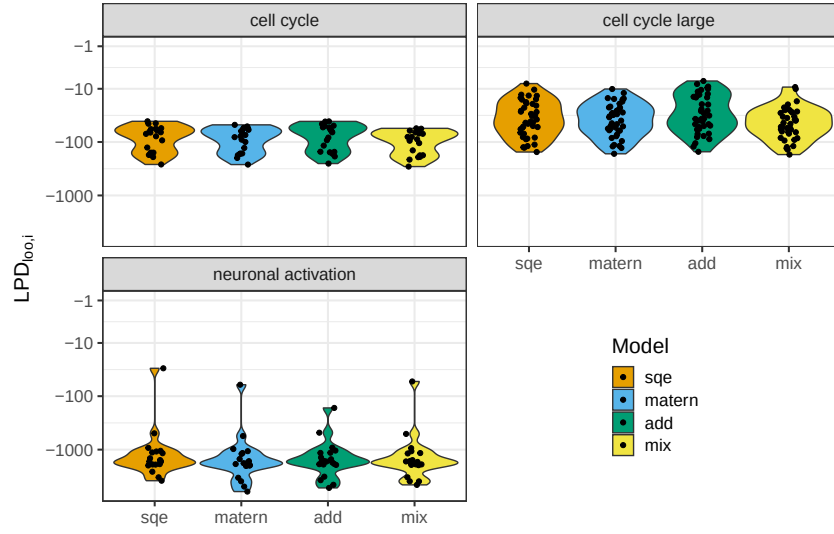


Figure 5.1: Log predictive density distributions for all models and two scRNAseq datasets

The LOOCVs were performed with 20 datapoints, except for *cell cycle large* where 40 datapoints were used.

more information to learn from and thus increase their predictive performance. Furthermore, the LPD values for the neuronal activation dataset are generally lower than for the cell cycle dataset. This indicates that the neuronal activation dataset has a low signal and a large variance in the gene expression and velocity values.

5.3 Conclusion

Under the shown conditions of the comparison of the predictive performance of the models, no significant difference between the models can be found. Thus, for now, it is recommended to pick the simplest of the proposed models, either the SQE or the Matérn 3/2 one. From the simulation study in Chapter 3 we concluded, that the SQE covariance function often is too smooth to model complex data. Furthermore, we showed that under the same fitting conditions, the simpler models achieve better convergence on the posterior distribution and higher effective sample sizes. Therefore, we recommend utilising the Matérn 3/2 covariance function for derivative multi-output GPs when modelling scRNAseq data. In addition, we demonstrated for the case study that

the additive covariance functions do not provide meaningful upgrades to the standard ones like the SQE and the Matérn 3/2 covariance function.

Chapter 6

Conclusion and Outlook

In this chapter, we will briefly collect all the results shown in the previous chapters before generally discussing the GP model and the approaches used to compare them. Lastly, we will give an outlook on future research needed to further elucidate the use of combination covariance functions in the context of modelling scRNAseq data.

6.1 Final Conclusion and Discussion

In the previous chapters, we investigated the use of combination covariance functions in derivative multi-output GPs for modelling scRNAseq data with RNA velocity and pseudotime estimates. In the simulation study shown in Chapter 2, we concluded that the standard SQE covariance function is unable to properly model complex dynamics. It models the presented data too smoothly and explains a large portion of the data’s variance as noise. Furthermore, we saw that the standard covariance functions result in better convergence and larger effective sample sizes compared to the additive combinations of covariance functions. From the LOOCV of the simulated dataset and the case study, we concluded that the additive covariance functions do not provide a substantial gain when modelling scRNAseq data. Based on all conclusions made here, the Matérn 3/2 covariance function is to be preferred when modelling scRNAseq data.

Considering the various methods we used to compare the different models, we can conclude that the simulation study gave extensive insight into the

fitting behaviours of the models. Even with the large mismatch of hyperparameters between the models, we were able to identify unwanted behaviours as was the case for the SQE covariance function. The LOOCVs in combination with the ELPD measure allowed for an extensive comparison of the predictive performances based on the whole predictive distribution of the models, fully utilising the Bayesian framework we used [Bürkner et al., 2021]. Finally, we can say that all methods we used to compare the models were useful and we recommend their utilisation for further analysis on the topic of combination covariance function for modelling scRNAseq data.

6.2 Outlook

Since we were unable to provide any combination covariance function that models scRNAseq data better than the standard covariance functions, we want to give an extensive outlook on what research needs to be done to find better covariance functions. First of all, in this thesis, we have not investigated the multiplicative combination in further detail due to a more complex derivative structure. This method of combining two covariance functions can produce highly complex and more interesting properties compared to the additive combinations [Duvenaud, 2014], making it important to investigate. Hence, it would be interesting to study this type of combination to find out if they add substantially more meaningful upgrades to the standard covariance functions as compared to the additive ones when modelling scRNAseq data.

Furthermore, we trained our Gaussian process models only on small datasets of 20 or 40 data points due to their bad runtime scaling. This number is unrealistically low compared to common dataset sizes of scRNAseq data, which can contain up to millions of cells [Jovic et al., 2022]. To incorporate larger datasets into the search for a suitable covariance function, one has to either adapt one of the various existing GP approximation methods to make them applicable to derivative GPs or develop new ones. Even when the full-sized scRNAseq dataset issue is solved from the GP side, the bad scaling of the leave-one-out cross-validation remains. However, one could approximate the exact LOOCV method with Pareto-smoothed importance sampling [Vehtari et al., 2017]. Hence, there are numerous potential avenues and

techniques that must be developed to further improve the comparison of combination covariance functions and achieve more definitive results.

Furthermore, the models can be fitted on more scRNAseq datasets to study their applicability to a wider range of datasets. This allows for the investigation of what needs to be developed on a case-by-case basis. Especially since the underlying biological processes can be diverse and produce considerably different dynamics. In the end, we can conclude that our work is the first step in the exploration of using combination covariance functions for modelling scRNAseq data including RNA velocity and pseudotime estimates, leaving plenty of room for future research.

Appendix A

Further Equations and Figures

A.1 Covariance Function Derivatives

A.1.1 Squared Exponential

Original covariance function

$$K(t_i, t_j) = \alpha^2 \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right)$$

First derivative (t_j)

$$\begin{aligned}\frac{\partial K(t_i, t_j)}{\partial t_j} &= \alpha^2 \frac{\partial}{\partial t_j} \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \\ &= \alpha^2 \cdot \frac{1}{2\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \cdot \frac{\partial}{\partial t_j} - (t_i - t_j)^2 \\ &= \alpha^2 \cdot \frac{1}{2\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \cdot 2(t_i - t_j) \\ &= \alpha^2 \frac{t_i - t_j}{\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right)\end{aligned}$$

First derivative (t_i)

$$\begin{aligned}
\frac{\partial K(t_i, t_j)}{\partial t_i} &= (\text{analogous to above}) \\
&= \alpha^2 \cdot \frac{1}{2\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \cdot (-2(t_i - t_j)) \\
&= -\alpha^2 \frac{t_i - t_j}{\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right)
\end{aligned}$$

Second derivative

$$\begin{aligned}
\frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} &= \frac{\partial}{\partial t_j} - \alpha^2 \frac{t_i - t_j}{\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \\
&= \left(\exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \cdot \frac{\partial}{\partial t_j} - \alpha^2 \frac{t_i - t_j}{\rho^2} \right) \\
&\quad + \left(-\alpha^2 \frac{t_i - t_j}{\rho^2} \cdot \frac{\partial}{\partial t_j} \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \right) \\
&= \left(\exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \cdot \frac{\alpha^2}{\rho^2} \right) \\
&\quad + \left(\alpha^2 \frac{-(t_i - t_j)^2}{\rho^2} \cdot \frac{t_i - t_j}{\rho^2} \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \right) \\
&= \left(\exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \cdot \frac{\alpha^2}{\rho^2} \right) - \left(\alpha^2 \frac{(t_i - t_j)^2}{\rho^4} \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \right) \\
&= \left(\frac{\alpha^2}{\rho^2} - \frac{\alpha^2 (t_i - t_j)^2}{\rho^4} \right) \cdot \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right) \\
&= \frac{\alpha^2}{\rho^4} \cdot (\rho^2 - (t_i - t_j)^2) \exp\left(-\frac{(t_i - t_j)^2}{2\rho^2}\right)
\end{aligned}$$

A.1.2 Matérn 3/2**Original covariance function**

$$K(t_i, t_j) = \alpha^2 \cdot \left(1 + \frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \cdot \exp\left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho}\right)$$

First derivative (t_j)

$$\begin{aligned}
\frac{\partial K(t_i, t_j)}{\partial t_j} &= \alpha^2 \frac{\partial}{\partial t_j} \left(1 + \frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \cdot \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \\
&= \alpha^2 \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \cdot \frac{\partial}{\partial t_i} \left(1 + \frac{\sqrt{3}}{\rho} |t_i - t_j| \right) \\
&\quad + \alpha^2 \frac{\partial}{\partial t_i} \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \cdot \left(1 + \frac{\sqrt{3}}{\rho} |t_i - t_j| \right) \\
&= \alpha^2 \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \cdot \frac{-\sqrt{3}(t_i - t_j)}{\rho |t_i - t_j|} \\
&\quad + \alpha^2 \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \frac{\sqrt{3}(t_i - t_j)}{\rho |t_i - t_j|} \cdot \left(1 + \frac{\sqrt{3}}{\rho} |t_i - t_j| \right) \\
&= \alpha^2 \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \\
&\quad \cdot \left(\frac{-\sqrt{3}(t_i - t_j)}{\rho |t_i - t_j|} + \left(\frac{\sqrt{3}(t_i - t_j)}{\rho |t_i - t_j|} \cdot \left(1 + \frac{\sqrt{3}}{\rho} |t_i - t_j| \right) \right) \right) \\
&= \alpha^2 \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \\
&\quad \cdot \frac{-\sqrt{3}(t_i - t_j)}{\rho |t_i - t_j|} \cdot \left(1 + \left(-1 \cdot \left(-1 + \frac{\sqrt{3}}{\rho} |t_i - t_j| \right) \right) \right) \\
&= \alpha^2 \exp \left(-\frac{\sqrt{3 \cdot (t_i - t_j)^2}}{\rho} \right) \cdot \frac{-\sqrt{3}(t_i - t_j)}{\rho |t_i - t_j|} \cdot \left(1 - 1 - \frac{\sqrt{3}}{\rho} |t_i - t_j| \right) \\
&= \alpha^2 \cdot \left(\frac{3(t_i - t_j)}{\rho^2} \right) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right)
\end{aligned}$$

First derivative (t_i)

$$\begin{aligned}
\frac{\partial K(t_i, t_j)}{\partial t_i} &= (\text{analogous to above}) \\
&= -\alpha^2 \cdot \left(\frac{3(t_i - t_j)}{\rho^2} \right) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right)
\end{aligned}$$

Second derivative

$$\begin{aligned}
\frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} &= \frac{\partial}{\partial t_j} \left(-\alpha^2 \cdot \left(\frac{3(t_i - t_j)}{\rho^2} \right) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right) \right) \\
&= -\alpha^2 \cdot \frac{3}{\rho^2} \cdot \frac{\partial}{\partial t_j} \left((t_i - t_j) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right) \right) \\
&= -\alpha^2 \frac{3}{\rho^2} \left(\frac{\partial t_i - t_j}{\partial t_j} \exp(\dots) + (t_i - t_j) \cdot \frac{\partial}{\partial t_j} \exp(\dots) \right) \\
&= -\alpha^2 \frac{3}{\rho^2} \left(-\exp(\dots) + (t_i - t_j) \cdot \frac{\sqrt{3}(t_i - t_j)}{\rho|t_i - t_j|} \exp(\dots) \right) \\
&= \alpha^2 \frac{3}{\rho^2} \left(\exp(\dots) - (t_i - t_j) \cdot \frac{\sqrt{3}(t_i - t_j)}{\rho|t_i - t_j|} \exp(\dots) \right) \\
&= \alpha^2 \frac{3}{\rho^2} \left(1 - (t_i - t_j) \cdot \frac{\sqrt{3}(t_i - t_j)}{\rho|t_i - t_j|} \right) \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right) \\
&= \alpha^2 \frac{3}{\rho^2} \left(1 - \frac{\sqrt{3}(t_i - t_j)^2}{\rho|t_i - t_j|} \right) \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right) \\
&= \alpha^2 \frac{3}{\rho^2} \left(1 - \frac{\sqrt{3} \cdot (t_i - t_j)^2 \sqrt{(t_i - t_j)^2}}{\rho|t_i - t_j|} \right) \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right) \\
&= \alpha^2 \frac{3}{\rho^2} \left(1 - \frac{\sqrt{3} \cdot (t_i - t_j)^2 |t_i - t_j|}{\rho|t_i - t_j|} \right) \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right) \\
&= \alpha^2 \cdot \frac{3}{\rho^2} \cdot \left(1 - \frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right) \cdot \exp \left(-\frac{\sqrt{3}(t_i - t_j)^2}{\rho} \right)
\end{aligned}$$

A.1.3 Combination Covariance Functions

Additive combination

$$\begin{aligned}
K(t_i, t_j) &= K_1(t_i, t_j) + K_2(t_i, t_j) \\
\frac{\partial K(t_i, t_j)}{\partial t_j} &= \frac{\partial K_1(t_i, t_j)}{\partial t_j} + \frac{\partial K_2(t_i, t_j)}{\partial t_j} \\
\frac{\partial K(t_i, t_j)}{\partial t_i} &= \frac{\partial K_1(t_i, t_j)}{\partial t_i} + \frac{\partial K_2(t_i, t_j)}{\partial t_i} \\
\frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} &= \frac{\partial^2 K_1(t_i, t_j)}{\partial t_i \partial t_j} + \frac{\partial^2 K_2(t_i, t_j)}{\partial t_i \partial t_j}
\end{aligned}$$

Mixing combination

$$\begin{aligned}
K(t_i, t_j) &= m \cdot K_1(t_i, t_j) + (1 - m) \cdot K_2(t_i, t_j) \\
\frac{\partial K(t_i, t_j)}{\partial t_j} &= m \cdot \frac{\partial K_1(t_i, t_j)}{\partial t_j} + (1 - m) \cdot \frac{\partial K_2(t_i, t_j)}{\partial t_j} \\
\frac{\partial K(t_i, t_j)}{\partial t_i} &= m \cdot \frac{\partial K_1(t_i, t_j)}{\partial t_i} + (1 - m) \cdot \frac{\partial K_2(t_i, t_j)}{\partial t_i} \\
\frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} &= m \cdot \frac{\partial^2 K_1(t_i, t_j)}{\partial t_i \partial t_j} + (1 - m) \cdot \frac{\partial^2 K_2(t_i, t_j)}{\partial t_i \partial t_j}
\end{aligned}$$

Multiplicative combination

$$\begin{aligned}
K(t_i, t_j) &= K_1(t_i, t_j) \cdot K_2(t_i, t_j) \\
\frac{\partial K(t_i, t_j)}{\partial t_j} &= \frac{\partial K_1(t_i, t_j)}{\partial t_j} \cdot K_2(t_i, t_j) + K_1(t_i, t_j) \cdot \frac{\partial K_2(t_i, t_j)}{\partial t_j} \\
\frac{\partial K(t_i, t_j)}{\partial t_i} &= \frac{\partial K_1(t_i, t_j)}{\partial t_i} \cdot K_2(t_i, t_j) + K_1(t_i, t_j) \cdot \frac{\partial K_2(t_i, t_j)}{\partial t_i} \\
\frac{\partial^2 K(t_i, t_j)}{\partial t_i \partial t_j} &= \frac{\partial}{\partial t_j} \left(\frac{\partial K_1(t_i, t_j)}{\partial t_i} \cdot K_2(t_i, t_j) \right) + \frac{\partial}{\partial t_j} \left(K_1(t_i, t_j) \cdot \frac{\partial K_2(t_i, t_j)}{\partial t_i} \right) \\
&= \frac{\partial^2 K_1(t_i, t_j)}{\partial t_i \partial t_j} \cdot K_2(t_i, t_j) + \frac{\partial K_1(t_i, t_j)}{\partial t_j} \cdot \frac{\partial K_2(t_i, t_j)}{\partial t_i} \\
&\quad + \frac{\partial K_1(t_i, t_j)}{\partial t_i} \cdot \frac{\partial K_2(t_i, t_j)}{\partial t_j} + K_1(t_i, t_j) \cdot \frac{\partial^2 K_2(t_i, t_j)}{\partial t_i \partial t_j}
\end{aligned}$$

A.2 Investigation of SQE Performance

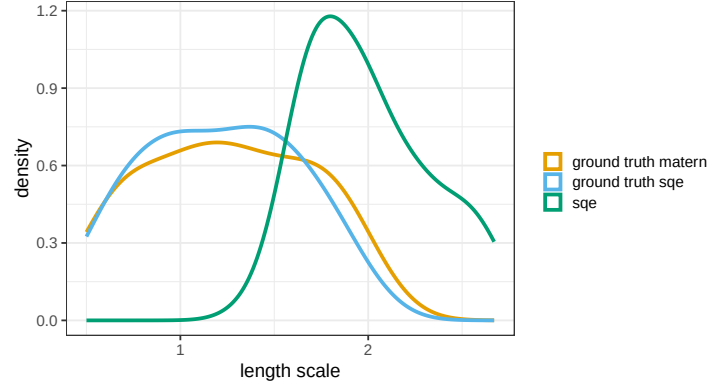


Figure A.1: Comparison of ground truth length scale values with posterior values of SQE model

For the SQE model the mean of all posterior samples per simulation study trial is shown.

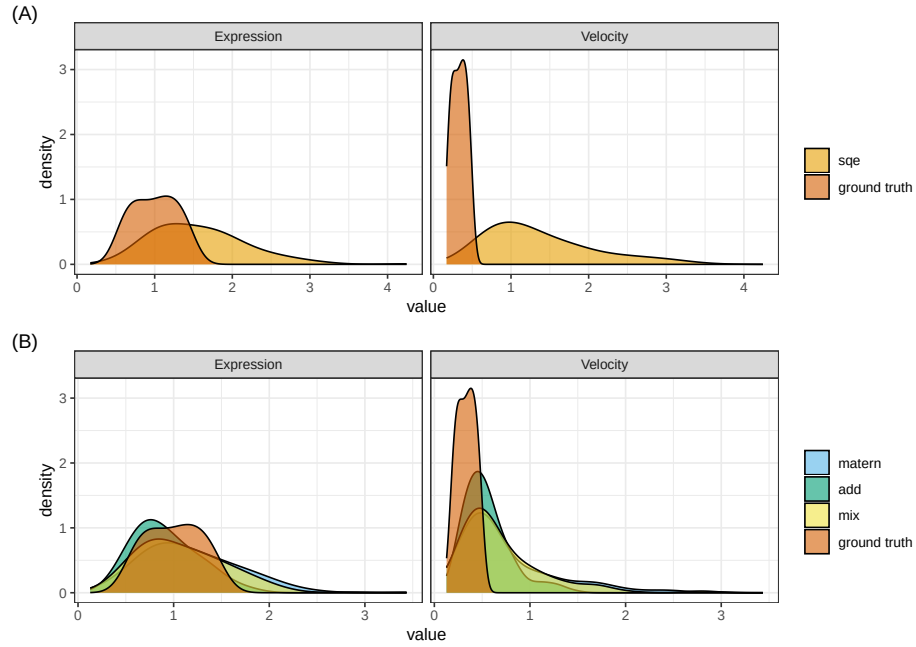


Figure A.2: Comparison of ground truth noise standard deviation values with values of (A) SQE model and (B) rest of the models

For the fitted model the mean of all posterior samples per simulation study trial is shown.

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Selbständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Masterarbeit selbständig und nur mit den angegebenen Hilfsmitteln angefertigt habe und dass alle Stellen, die dem Wortlaut oder dem Sinne nach anderen Werken entnommen sind, durch Angaben von Quellen als Entlehnung kenntlich gemacht worden sind. Diese Masterarbeit wurde in gleicher oder ähnlicher Form in keinem anderen Studiengang als Prüfungsleistung vorgelegt.

Ort, Datum

Unterschrift